Simultaneous Equations & Inequalities

Section 1: Linear & Non-Linear Simultaneous Systems

1 Solve the simultaneous equations $y = 2x - 3$ and $x^2 + y^2 = 26$. Which pair of points represents the solutions?

a. $(1, -1)$ and $\left(\frac{17}{5}, \frac{19}{5}\right)$

b. $(3, 3)$ and $(-1, -5)$

c. $(5, 7)$ and $(1, -1)$

d. $(2, 1)$ and $(-2, -7)$

2 The line $x + y = 5$ intersects the hyperbola $xy = 6$. Find the coordinates of the two intersection points.

a. $(2, 3)$ and $(3, 2)$

b. $(1, 6)$ and $(6, 1)$

c. $(2, 3)$ and $(-2, -3)$

d. $(4, 1)$ and $(1, 4)$

3 Find the points of intersection of the line $2x - y = 1$ and the curve $y = x^2 - 4$.

a. $(3, 5)$ and $(-1, -3)$

b. $(2, 3)$ and $(-2, -5)$

c. $(4, 7)$ and $(1, 1)$

d. $(5, 9)$ and $(0, -1)$

Section 2: Linear and Quadratic Inequalities

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4 Solve the quadratic inequality $x^2 - 2x - 8 < 0$.

a. $-2 < x < 4$

b. $x < -2$ or $x > 4$

c. $-4 < x < 2$

d. $x < -4$ or $x > 2$

5 Find the set of values of x satisfying $2x^2 + 5x - 3 \geq 0$.

a. $x \leq -3$ or $x \geq \frac{1}{2}$

b. $-3 \leq x \leq \frac{1}{2}$

c. $x \leq -\frac{1}{2}$ or $x \geq 3$

d. $-1 \leq x \leq 3$

6 Solve the simultaneous inequalities $3x - 5 < 7$ and $x^2 - 9 \geq 0$.

a. $x \leq -3$ or $3 \leq x < 4$

b. $-3 \leq x \leq 3$

c. $x < 4$

d. $3 \leq x < 4$

Section 3: Geometrical Applications & Line-Curve Systems

7 For what range of values of m does the line $y = mx$ not intersect the curve $y = x^2 + 9$?

a. $-6 < m < 6$

b. $m < -6$ or $m > 6$

c. $-3 < m < 3$

d. $m > 6$

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8 The perimeter of a rectangle is 28 cm and its area is at least 45 cm^2. What is the range of possible values for the width w ($w \le l$)?

a. $5 \le w \le 7$

b. $4 \le w \le 9$

c. $3 \le w \le 7$

d. $w \ge 5$

9 Find the coordinates of the points where the line $y = 3x - 2$ is normal to the curve $y = x^2 - 4x + 3$.

a. $(1, 0)$

b. $(2, -1)$

c. $(0, -2)$

d. $(3, 0)$

Section 4: Advanced Problem Solving & Exam Challenges

10 Find the range of values of k for which the line $y = kx - 1$ intersects the circle $x^2 + y^2 = 4$ in two distinct points.

a. $k^2 > -\frac{3}{4}$ (all real k)

b. $k > \sqrt{3}$

c. $k < -\sqrt{3}$ or $k > \sqrt{3}$

d. $-1 < k < 1$

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11 A company finds that its profit P (in thousands of dollars) satisfies $P(x) = -2x^2 + 28x - 80 \geq 10$, where x is the production level in hundreds of units. What is the valid production interval for x?

a. $5 \leq x \leq 9$

b. $4 \leq x \leq 10$

c. $3 \leq x \leq 11$

d. $6 \leq x \leq 8$

12 Find all values of x satisfying the rational inequality $\dfrac{x-3}{x+2} < 0$.

a. $-2 < x < 3$

b. $x < -2$ or $x > 3$

c. $x < 3$

d. $-3 < x < 2$

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